

Disclaimer: In accordance with the assessment instructions, the Data Request presented in this section was generated with the assistance of Generative AI. The variables requested were informed by the identified STADIOalot search-relevance problem, the available data sources described in the client information pack, and the requirements of the proposed Data Science investigation. The AI-generated output was reviewed and refined to ensure that the requested variables are relevant to the proposed project.

Data Request Introduction

This proposed research investigates whether customer search behaviour and product metadata from sellers can be used to predict product relevance and support more relevant search ranking on STADIOalot. Therefore two sources of data are required.

The first data request focuses on **Product Catalogue and Seller Data**, which describes the attributes of products available on the Stadioalot platform and how the products are represented . This data is required to determine how well a product corresponds with a customer's search query.

The second data request focuses on **Customer Search and Behavioural Data**, which captures what customers search for and how they interact with the products presented to them. This includes search queries, product impressions, clicks, basket additions, purchases and search abandonment.

Both datasets would ultimately be used to link customer search intent and product relevance.

Table 1 — Seller & Product Data Request

Column name	Data type	Example	Additional information / purpose
product_id	String	SKU12345	Unique identifier for each product; used to join product and behavioural data.
seller_id	String	SELLER102	Anonymised identifier for the seller listing the product.
product_title	String/Text	Nike Women's Running Shoe	Seller-provided product title; important for matching products to search queries.

product_description	String/Text	Lightweight running shoe...	Product description that may contain terms relevant to the customer's search.
product_category	Categorical	Footwear	Main category assigned to the product.
product_subcategory	Categorical	Running Shoes	More specific product classification.
brand	Categorical/String	Nike	Brand associated with the product; may form part of customer search intent.
product_tags	String/List	running; women; sports	Search/indexing terms associated with the product.
product_attributes	String/JSON	Colour: Black; Size: 6	Product characteristics that can be compared with query terms.
colour	Categorical	Black	Specific product attribute commonly included in searches.
size	Categorical	6	Product size, where applicable.
product_price	Decimal/Float	1299.99	Product price at the time of the search.
discount_percentage	Float	15.0	Indicates whether/how heavily the product is discounted.
inventory_status	Categorical	In Stock	Indicates whether the product is currently available for purchase.
inventory_quantity	Integer	37	Number of units currently available.
distribution_centre	Categorical/String	JHB_DC01	Location where inventory is held; potentially relevant to availability/delivery.

<code>estimated_delivery_days</code>	Integer	2	Expected fulfilment time for the product.
<code>seller_rating</code>	Float	4.6	Seller performance/quality indicator.
<code>product_rating</code>	Float	4.4	Average customer rating for the product.
<code>review_count</code>	Integer	286	Number of reviews contributing to the product rating.

Table 2 — Customer Search & Behavioural Data Request

Column name	Data type	Example	Additional information / purpose
<code>customer_id</code>	String	CUST_9281	Anonymised customer identifier used to connect interactions without directly identifying the customer.
<code>session_id</code>	String	SES_78542	Connects searches, displayed products and subsequent actions within one browsing session.
<code>search_id</code>	String	SEARCH_45892	Unique identifier for an individual search event.
<code>search_query</code>	String/Text	black running shoes	Exact search terms entered by the customer.
<code>search_timestamp</code>	DateTime	2026-06-15 14:32	Date and time the search occurred; useful for temporal/seasonal patterns.
<code>device_type</code>	Categorical	Mobile	Device used during the search.

search_result_product_id	String	SKU12345	Product shown for the customer's query; links behavioural and product data.
search_result_position	Integer	4	Position in which the product appeared in the search results.
sponsored_listing	Boolean	True	Whether the displayed result was a paid/sponsored placement.
product_impression	Boolean	True	Whether the product was actually displayed to the customer.
product_clicked	Boolean	True	Whether the customer clicked the displayed product.
click_timestamp	DateTime	2026-06-15 14:33	Time at which the product was clicked.
added_to_basket	Boolean	True	Whether the product was added to the customer's basket.
basket_timestamp	DateTime	2026-06-15 14:35	When the product was added to basket.
purchased	Boolean	True	Whether the interaction eventually resulted in a purchase.
purchase_timestamp	DateTime	2026-06-15 14:42	When the purchase occurred.
quantity_purchased	Integer	1	Number of units purchased following the interaction.
search_abandoned	Boolean	False	Whether the customer left the search without meaningful interaction/conversion.
query_reformulated	Boolean	True	Whether the customer changed/re-entered their query after seeing the results.
previous_search_query	String/Text	running shoes	Previous query in the session; can help identify how customers refine their search intent.